

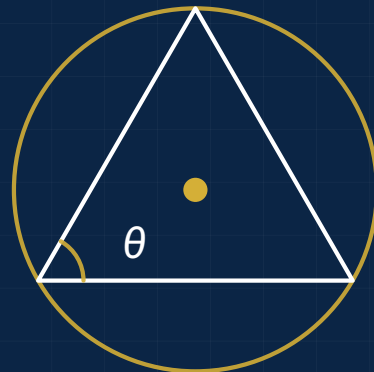


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CHAPTER 4 • GEOMETRY & MEASURES • TOPIC 1

Standard & Compound Units

Speed, density and pressure are one quantity divided by another. Cover the one you want in the triangle, match the units, then calculate.



5

VARIANTS

32

PAST-PAPER PARTS

45

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Converting units

V2 m/s ↔ km/h

V3 Speed, distance, time

V4 Density, mass, volume

V5 Pressure, force, area

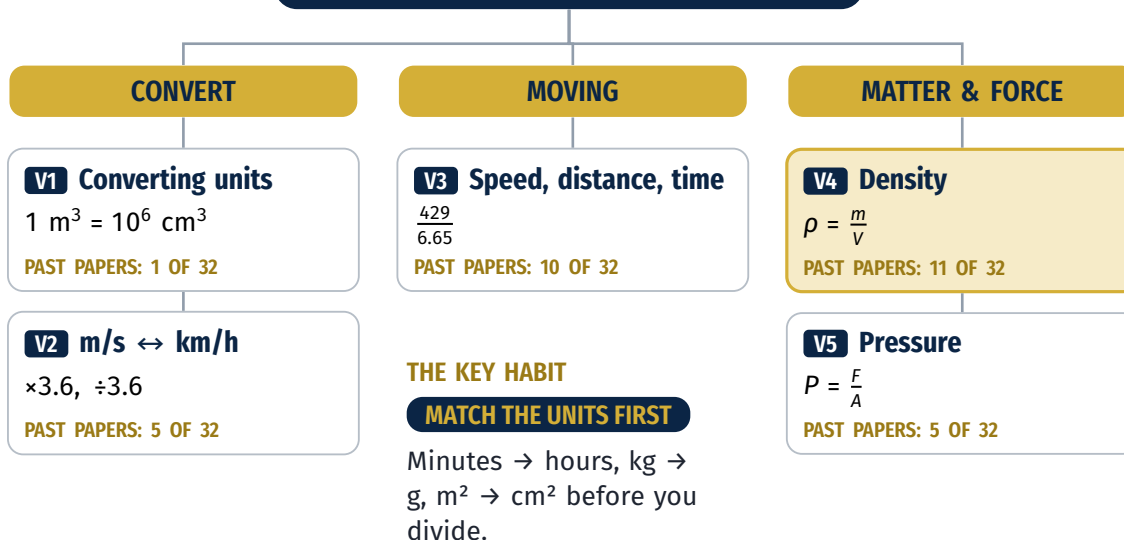
TOPIC MAP • EVERY VARIANT ON ONE PAGE

Units at a glance



Your notes

Standard & Compound Units



★ How often does each variant come up?

31 past-paper questions (Jan 2020 $\rightarrow$ Nov 2025) hold 32 parts; 45 practice parts — grouped by type.

	PAST PAPERS (OUT OF 32)	PRACTICE PARTS
V1 Converting units	1	8
V2 m/s and km/h	5	0
V3 Speed, distance, time	10	17
V4 Density, mass, volume	11	15
V5 Pressure, force, area	5	5

WHAT THIS TELLS YOU On every paper, usually twice, early (Q1–Q13, 2–5 marks). The maths is one division; the marks are lost on units — hours and minutes, g and kg, m and cm.

BEFORE YOU START

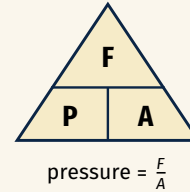
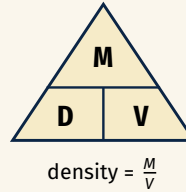
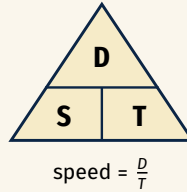
The big idea



Your notes
A compound unit is “per”.

km/h = km **per** hour, g/cm³ = g **per** cm³, N/m² = N **per** m². The unit tells you the division.

Cover what you want in the triangle: what's left is the calculation.

**Time is not decimal****Minutes → hours**

39 min = $\frac{39}{60}$ =
0.65 h
Divide by 60.

h min → hours

6 h 39 min =
6.65 h
Not 6.39!

Hours → h min

0.4 h = 24 min
Decimal part ×60.

Hours → seconds

×3600
60 × 60.

**Area and volume units****Length**

1 m = 100 cm
×100.

Area

1 m² = 100² =
10 000 cm²
Square the factor.

Volume

1 m³ = 100³ = 10⁶ cm³
Cube the factor.

**CALCULATOR FIRST, THEN CHECK**

Calculator: use the degrees–minutes key for times: 6°39' ÷ ... , or type 6 + 39 ÷ 60. Then check the answer's size makes sense.

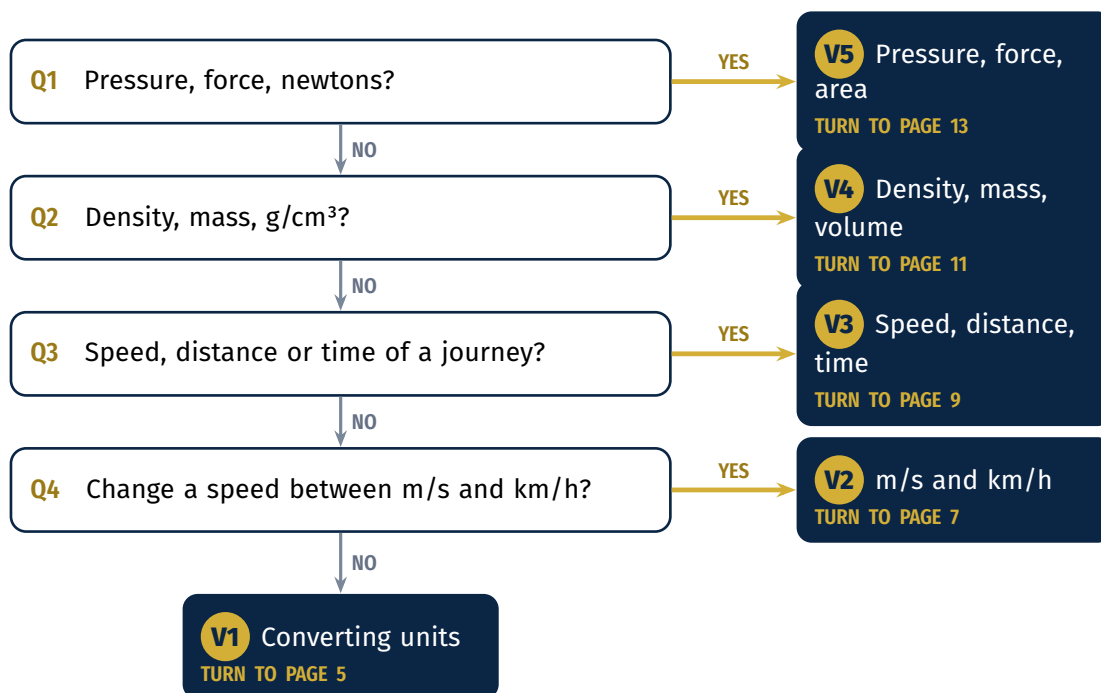
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant – and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of “How do I do it?” each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Converting Units

Length: the factor. Area: factor squared.

Volume: factor cubed.

SEEN IN PAST PAPERS

1 / 32

PRACTICE QUESTIONS: 8



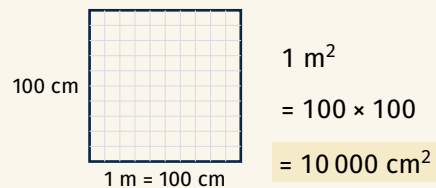
Your notes

Came up in: Jan 2020 P2H Q2

? What is it?

In plain words

1 m = 100 cm, but a 1 m by 1 m square is $100 \times 100 = 10\,000 \text{ cm}^2$, and a 1 m cube is $100 \times 100 \times 100 = 1\,000\,000 \text{ cm}^3$.

Big unit → small unit: **multiply**.Small → big: **divide**.

IT LOOKS LIKE

$32.4 \text{ m}^3 \rightarrow \text{cm}^3$

$4 \text{ m}^2 \rightarrow \text{cm}^2$

How do I do it?

1 Factor

The length factor: m → cm 100, cm → mm 10, km → m 1000.

2 Power

Area: square it. Volume: cube it.

3 Direction

To a smaller unit: ×. To a bigger unit: ÷.

4 Sense

More small units than big ones.

SQUARE FOR AREA, CUBE FOR VOLUME

$1 \text{ m}^3 = 10^6 \text{ cm}^3$

V1 Converting Units

VARIANT 1 · SEE IT DONE, THEN TRY


 Your notes
 WHEN YOU SEE → YOU DO

"Change 32.4 m^3 into cm^3 " → $\times 100^3 = \times 1\,000\,000$

"Change 4 m^2 into cm^2 " → $\times 100^2 = \times 10\,000$

A time in minutes for a km/h answer → $\div 60$ **to hours**

**WORKED EXAMPLE** **PAST PAPER**

January 2020 · Paper 2H · Q2 · 2 marks

Change 32.4 m^3 into cm^3 .

STEP 1 $1 \text{ m} = 100 \text{ cm}$ ◀ length factor

STEP 2 $1 \text{ m}^3 = 100^3 = 1\,000\,000 \text{ cm}^3$ ◀ cube it

STEP 3 $32.4 \times 1\,000\,000$ ◀ bigger to smaller: $\times$

STEP 4 $= 32\,400\,000$ ◀

ANSWER $32\,400\,000 \text{ cm}^3$ ✓ 3.24×10^7 ✓**WATCH OUT** $\times 32.4 \times 100 = 3240$ ✓ $\times 100^3$

Volume needs the factor cubed.

**EXAM TIP**

Draw a 1 m cube and write 100 cm on each edge — the cube is obvious.

**Check yourself**

calculator first · answers on page 15

1.1

Practice · Easy

Change 4 m^2 into cm^2 .

Answer:

1.2

Practice · Easy

Change 15 cm^3 into mm^3 .

Answer:

1.3

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Change 3.5 m^2 into cm^2 .

Answer:

V2

VARIANT 2

Converting m/s and km/h

m/s $\rightarrow$ km/h: $\times 3600 \div 1000 = \times 3.6$. km/h $\rightarrow$ m/s: $\div 3.6$.

SEEN IN PAST PAPERS

5 / 32

PRACTICE QUESTIONS: 0


 Your notes

Came up in: Jan 2020 P2H Q10 • Jun 2022 P2H Q7 • Jun 2023 P2H Q9 • Jun 2024 P2H Q6 • Jun 2025 P1HR Q7

? What is it?

In plain words

Change **both parts** of the unit:
 metres $\leftrightarrow$ kilometres (1000) and
 seconds $\leftrightarrow$ hours (3600).

1 m/s = 3600 m per hour = **3.6 km/h**.
 So m/s $\times 3.6$ gives km/h.



IT LOOKS LIKE

50 m/s

81 km/h

50x m/s

How do I do it?

1 Distance

m $\rightarrow$ km: $\div 1000$ (or km $\rightarrow$ m: $\times 1000$).

2 Time

per second $\rightarrow$ per hour: $\times 3600$.

3 Combine

Net factor 3.6.

4 Sense

km/h number is bigger than the m/s number.

$\times 3.6$ TO km/h Show the 3600 and 1000 for full marks.

V2 Converting m/s and km/h

VARIANT 2 · SEE IT DONE, THEN TRY

 Your notes
 WHEN YOU SEE → YOU DO

"Change 50 metres per second to kilometres per hour" → $\times 3600, \div 1000$

"Change 81 km/h to m/s" → $\times 1000, \div 3600$

"w metres per second ... in terms of w" → $3.6w = \frac{18w}{5}$

 WORKED EXAMPLE **PAST PAPER**

June 2022 · Paper 2H · Q7 · 3 marks

Change a speed of 81 kilometres per hour to a speed in metres per second.

STEP 1 81 km/h = 81 000 m per hour ◀ km → m

STEP 2 81 000 ÷ 3600 ◀ per hour → per second

STEP 3 = 22.5 ◀

ANSWER 22.5 m/s

✓ $22.5 \times 3.6 = 81$ ✓

 WATCH OUT

✗ $81 \times 3.6 = 291.6$ m/s

✓ $81 \div 3.6 = 22.5$

km/h to m/s makes the number smaller.

 EXAM TIP

3 marks: write the $\times 1000$ and the $\div 3600$ as separate steps.

 Check yourself

calculator first · answers on page 15

2.1 Jan 2020 P2H

Change 50 m/s to km/h.

Answer:

2.2 Jun 2023 P2H

Change 27 km/h to m/s.

Answer:

2.3 Jun 2025 P1HR

Change 50x m/s to km/h.

Answer:

V3**VARIANT 3****Speed, Distance, Time**

Time in hours first, then $S = \frac{D}{T}$, $D = ST$ or $T = \frac{D}{S}$.

SEEN IN PAST PAPERS

10 / 32

PRACTICE QUESTIONS: 17



Your notes

Came up in: Jan 2020 P2HR Q5 • Oct/Nov 2020 P1HR Q7 • Oct/Nov 2020 P1H Q6 • Jan 2021 P1HR Q4 • Jan 2021 P2H Q1 • May 2021 P1H Q1 • Jan 2022 P1HR Q7 • Jun 2022 P2H Q3 • Jun 2023 P2HR Q3 • Jun 2024 P2H Q3

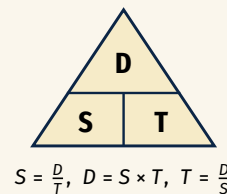
? What is it?**In plain words**

Change the time to

hours as a decimal: 6 h 39 min
 $= 6 + \frac{39}{60} = 6.65$ h. Then use the triangle.

Average speed for several stages

total distance
 $= \frac{\text{total distance}}{\text{total time}}$ — never the average of the speeds.

**IT LOOKS LIKE**

429 km in 6 h 39 min

three stages

How do I do it?**1 Time**To hours: minutes $\div 60$.**2 Triangle**

Cover what you want.

3 Several stagesTotal distance $\div$ total time.**4 Round**

As asked; give units.

MINUTES $\div 60$

6 h 39 min is not 6.39 h.

V3 Speed, Distance, Time

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"takes 6 hours 39 minutes ... work out the average speed" → $6 + \frac{39}{60}$ h, then $D \div T$

Journey in stages with different speeds → **Total distance ÷ total time**

"Give your answer in seconds" → **Hours × 3600**

WORKED EXAMPLE PAST PAPER

January 2021 • Paper 2H • Q1 • 3 marks

A train takes 6 hours 39 minutes to travel 429 km. Work out the average speed in km/h, correct to one decimal place.

STEP 1 $39 \div 60 = 0.65$ ◀ minutes to hours

STEP 2 $T = 6.65$ h ◀ total time

STEP 3 $S = \frac{429}{6.65} = 64.51 \dots$ ◀ distance ÷ time

STEP 4 ≈ 64.5 ◀ 1 d.p.

ANSWER 64.5 km/h✓ $64.5 \times 6.65 \approx 429$ ✓**WATCH OUT**✗ $429 \div 6.39 = 67.1$ ✓ $429 \div 6.65 = 64.5$

39 minutes is 0.65 of an hour, not 0.39.

EXAM TIP

"Write down all the stages": show the 6.65 line — it is a method mark.

Check yourself

calculator first • answers on page 15

3.1 Jun 2022 P2H

A plane flies 3980 km in 5 hours 24 minutes. Average speed to the nearest whole number?

Answer:

3.2 Jan 2022 P1HR

245 km in 2.5 h; 220 km at 80 km/h; 50 min at 72 km/h. Average speed for the whole journey (1 d.p.)?

Answer:

3.3 Oct/Nov 2020 P1H

A rocket travels 100 km at 28 440 km/h. How long, to the nearest second?

Answer:

V4**VARIANT 4** MOST COMMON IN THE EXAM

Density, Mass, Volume

Density = $\frac{\text{mass}}{\text{volume}}$: match grams with cm^3 ,
then cover what you need.

SEEN IN PAST PAPERS

11 / 32

PRACTICE QUESTIONS: 15



Your notes

Came up in: Jan 2020 P1HR Q9 • Jan 2020 P2H Q9 • Nov 2021 P1H Q10 • Jan 2022 P2HR Q21 • Jun 2022 P1HR Q7 • May 2023 P1HR

Q11 • May 2024 P1HR Q6 • Jun 2025 P2HR Q7 • Nov 2025 P2H Q26

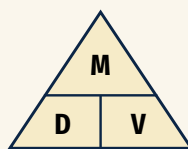
? What is it?

In plain words

Density tells you the mass of **1 cm^3** .

Gold 19.32 g/cm^3 : every cm^3 weighs
19.32 g.

Volumes come from the shape
(cuboid, cylinder, cone ...). Units
must match: $\text{kg} \rightarrow \text{g}$ when density is
in g/cm^3 .



$$D = \frac{M}{V}, M = D \times V, V = \frac{M}{D}$$

IT LOOKS LIKE

gold blocks

find w

which metal?



How do I do it?

1 Volume

From the shape's formula.

2 Units

 $\text{kg} \rightarrow \text{g}$, $\text{m}^3 \rightarrow \text{cm}^3$ to match the density.

3 Triangle

Cover what you want.

4 Context

“Least number of blocks”: round up.

g WITH cm^3

The most common variant here.

V4 Density, Mass, Volume

VARIANT 4 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"The mass of the statue is 5.73 kg;
density 19.32 g/cm³" → **5730 g first**

"Work out the least number of blocks" → **Divide, then round up**

"Determine the metal used" → **Find the density; match the table**

WORKED EXAMPLE PAST PAPER

January 2020 • Paper 1HR • Q9 • 5 marks

Gold blocks are cuboids 1.5 cm by 2 cm by 8 cm. A statue of mass 5.73 kg is made from them; gold has density 19.32 g/cm³. Work out the least number of blocks melted down.

STEP 1 $V = 1.5 \times 2 \times 8 = 24 \text{ cm}^3$ ◀ one block

STEP 2 $M = 19.32 \times 24 = 463.68 \text{ g}$ ◀ mass of one block

STEP 3 $5.73 \text{ kg} = 5730 \text{ g}; 5730 \div 463.68 = 12.36$ ◀ blocks needed

STEP 4 $\Rightarrow 13$ ◀ round up

ANSWER 13 blocks

✓ 12 blocks are only 5564 g ✓

WATCH OUT

✗ 12 blocks (rounded to nearest)

✓ 13 blocks

12 blocks aren't enough gold — round up.

EXAM TIP

Write the unit on every line: g, cm³,
g/cm³ — mismatches jump out.

Check yourself

calculator first • answers on page 15

4.1 Jan 2020 P2H

Gold has density 19.3 g/cm³. A gold bar has volume 150 cm³. Mass?

Answer:

4.2 May 2023 P1HR

An iron cuboid is w by 5 by 4 cm; density 7.8 g/cm³; mass 1950 g. Find w .

Answer:

4.3 Jun 2022 P1HR

An iron cylinder has diameter 18 cm, height 3.5 cm, mass 7.04 kg. Density (2 s.f.)?

Answer:

V5

VARIANT 5

Pressure, Force, Area

Pressure = $\frac{\text{force}}{\text{area}}$: the area is the face touching the ground.

SEEN IN PAST PAPERS

5 / 32

PRACTICE QUESTIONS: 5


 Your notes

Came up in: Jan 2020 P1H Q13 • Jan 2020 P2HR Q9 • Oct/Nov 2020 P1HR Q9 • Jun 2022 P1H Q7 • May 2024 P1H Q8

? What is it?

In plain words

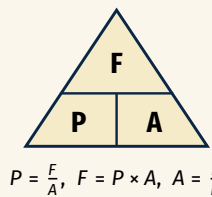
Pressure is the force on each unit of area: N/cm^2 or N/m^2 .

The **area** is the face **in contact** with the table. Find it from the pressure, then use it for lengths or volume.

IT LOOKS LIKE

cube on a table

cylinder on the ground



How do I do it?

1 Area

Contact face: square, rectangle, circle.

2 Units

N/cm^2 with cm^2 , N/m^2 with m^2 .

3 Triangle

Cover what you want.

4 Next

Area → side → volume if asked.

CONTACT FACE ONLY Not the total surface area.

V5 Pressure, Force, Area

VARIANT 5 • SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

"The cube exerts a force of 56 N;
pressure 0.14 N/cm². Volume?"

→ **Area = $\frac{F}{P}$, side, then cube**

"The face in contact is a square of side
1.5 m"

→ **A = 2.25 m²**

"Volume 1575 cm³, height 21 cm;
pressure?"

→ **Area = $\frac{V}{H}$ first**

**WORKED EXAMPLE PAST PAPER**

January 2020 • Paper 1H • Q13 • 4 marks

A cube of side w cm stands on a table. It exerts a force of 56 N; the pressure on the table is 0.14 N/cm². Work out the volume of the cube.

STEP 1 $A = \frac{F}{P} = \frac{56}{0.14} = 400 \text{ cm}^2$ ◀ contact area

STEP 2 $w^2 = 400 \Rightarrow w = 20$ ◀ side of the square face

STEP 3 $V = w^3 = 20^3$ ◀ cube

STEP 4 $= 8000$ ◀

ANSWER 8000 cm³

✓ $\frac{56}{400} = 0.14$ ✓

**WATCH OUT**

✗ $A = 56 \times 0.14$

✓ $A = \frac{56}{0.14}$

Cover A in the triangle: F over P .

**EXAM TIP**

Check the answer by recomputing the pressure from your area.

**Check yourself**

calculator first • answers on page 15

5.1

May 2024 P1H

A cylinder: volume 1575 cm³, height 21 cm, force 84 N. Work out the pressure on the table.

Answer:

5.2

Oct/Nov 2020 P1HR

The contact face is a square of side 1.5 m; pressure 34.8 N/m². Work out the force.

Answer:

5.3

Jan 2020 P2HR

Contact face 1.2 m by x m; force 27 N; pressure 30 N/m². Find x .

Answer:

CHECK YOURSELF

Answers



Every answer below was checked with sympy (check_answers.py).

1.1 $40\,000\text{ cm}^2$

1.2 $15\,000\text{ mm}^3$

1.3 $35\,000\text{ cm}^2$

2.1 180 km/h

2.2 7.5 m/s

2.3 $180x\text{ km/h}$

3.1 737 km/h

3.2 86.3 km/h

3.3 13 seconds

4.1 2895 g

4.2 $w = 12.5$

4.3 7.9 g/cm^3

5.1 1.12 N/cm^2

5.2 78.3 N

5.3 $x = 0.75$

THE WHOLE TOPIC ON ONE PAGE

Units cheat sheet



Your notes
V1 Converting 1/32

$$1 \text{ m}^3 = 10^6 \text{ cm}^3$$

Square for area, cube for volume.

V2 m/s ↔ km/h 5/32

$$\times 3.6$$

$$\times 3600 \div 1000.$$

V3 Speed 10/32

$$\frac{429}{6.65} = 64.5$$

Minutes ÷ 60; totals for averages.

V4 Density 11/32

$$D = \frac{M}{V}$$

g with cm³; round up blocks.**V5 Pressure 5/32**

$$P = \frac{F}{A}$$

Contact face only.

**TOP 3 LOST MARKS**

1. 6 h 39 min as 6.39 ×

2. kg with g/cm³ ×

3. ×100 for an area or volume ×

BOTTOM LINE Match the units • cover the triangle • minutes ÷ 60 • square for area, cube for volume.